

E1	Understand and use the definitions of sine, cosine and tangent for all arguments; the sine and cosine rules; the area of a triangle in the form $\frac{1}{2}ab\sin C$ Work with radian measure, including use for arc length and area of sector.
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E2	Understand and use the standard small angle approximations of sine, cosine and tangent $\sin \theta \approx \theta$, $\cos \theta \approx 1 - \frac{\theta^2}{2}$, $\tan \theta \approx \theta$ where θ is in radians.

	Content
E3	Understand and use the sine, cosine and tangent functions; their graphs, symmetries and periodicity. Know and use exact values of sin and cos for 0 , $\frac{\pi}{6}$, $\frac{\pi}{4}$, $\frac{\pi}{3}$, $\frac{\pi}{2}$, π and multiples thereof, and exact values of tan for 0 , $\frac{\pi}{6}$, $\frac{\pi}{4}$, $\frac{\pi}{3}$, π and multiples thereof.

	Content
E4	Understand and use the definitions of secant, cosecant and cotangent and of arcsin, arccos and arctan; their relationships to sine, cosine and tangent; understanding of their graphs; their ranges and domains.

	Content
E5	Understand and use $\tan \theta \equiv \frac{\sin \theta}{\cos \theta}$ Understand and use $\sin^2 \theta + \cos^2 \theta \equiv 1$; $\sec^2 \theta \equiv 1 + \tan^2 \theta$ and $\operatorname{cosec}^2 \theta \equiv 1 + \cot^2 \theta$

	Content
E6	Understand and use double angle formulae; use of formulae for $\sin(A \pm B)$, $\cos(A \pm B)$ and $\tan(A \pm B)$; understand geometrical proofs of these formulae. Understand and use expressions for $a \cos \theta + b \sin \theta$ in the equivalent forms of $r \cos(\theta \pm \alpha)$ or $r \sin(\theta \pm \alpha)$

E7	Solve simple trigonometric equations in a given interval, including quadratic equations in $\sin$, $\cos$ and $\tan$ and equations involving multiples of the unknown angle.
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E8	Construct proofs involving trigonometric functions and identities.

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E9	Use trigonometric functions to solve problems in context, including problems involving vectors, kinematics and forces.